PG4200-H: Data Structures and Algorithms
Prof. Dr. Rashmi Gupta
Kristiania University College, Oslo, Norway

Final Exam

- This exam includes a semester assignment in a group (2 students).
- This means that you are being provided with an assignment and you are expected to submit the below-mentioned documentation and a video file in a Zip file format:
  - **Assignment code scripts**
  - **10-minute video demonstration of code and explanation of the knowledge gained for each of the asked questions.**
  - **Report** that explains:
    - Your solutions in an algorithmic way with the relevant code snippet. Carefully explain your solutions to the problems in the text as well.
- **Duration:** 4 weeks.
- **Grading scale:** The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade and F is fail
- **Weighting:** 100% of the overall grade
- Start uploading your exam paper/file ahead of time, as it may take a long time to upload.
- Exam papers that are not handed in on Wiseflow by the specified time of the submission date will not proceed to assessment. No late solutions will be accepted. Your submitted documents will be checked for plagiarism.
- Before submitting, remember to check that all files can be opened and that every file is included. It may be a good idea to check the saved files on several machines before submitting them in Wiseflow.
- Incorrect file format or lacking documents may result in the submission not being passed or assessed.
- Show your work, and be creative, as partial credit will be given. You will be graded not only on the correctness of your answer but also on the clarity with which you express it.

Attachment: Grading Scale

The table below shows how the Universities Norway council (UHR) defines general, qualitative requirements for the corresponding letter grades. The criteria below should be used as a guideline in assigning grades for examinations unless other assessment criteria are explicitly provided for a particular assignment.

Letter grade	Descriptor	General, not subject-specific, description of assessment criteria
A	Outstanding	**Outstanding performance that clearly stands out.** The student has extremely good knowledge of the usage of different Data Structures based on a specific given problem.
B	Very good	**Very strong performance.** The candidate displays good knowledge of learning outcomes.
C	Good	**A good performance that satisfies most assessment criteria.** The candidate displays a good knowledge in the most important areas of assessment.
D	Fair	**An acceptable performance with some clear deficiencies**. The candidate somewhat displays a level of sound knowledge of learnt DSA concepts.
E	Sufficient	**A performance that only satisfies the minimum requirements.** The candidate displays a poor level of knowledge of learnt DSA concepts.
F	Fail	**A performance that does not satisfy the most basic formal requirements.** The candidate lacks sound knowledge of learnt DSA concepts.

Point total and grading scale [1]

Letter Grade	Corresponding numerical grade	Westerdals Grading scale
A	90-100	93 -100
B	80 – 89	78 – 92
C	60 - 79	59 - 77
D	50 - 59	51 - 58
E	40 - 49	40 - 50
F	0 - 39	0 - 39

Good luck!

In this practice, you will use the **Wine Quality Dataset**, which can be downloaded for free from: https://archive.ics.uci.edu/ml/datasets/wine+quality

Problem 1. BubbleSort (25 Marks)
Initially, use the **unique alcohol content values** of each wine from the combined red wine and white wine .csv files.

a) Implement both an **optimised** and a **non-optimised** Bubble Sort algorithm so that all alcohol content values are arranged in ascending order.
b) Calculate the **time complexity** for sorting the dataset. Does it change if you randomly shuffle the list before sorting? Why or why not?

Problem 2. InsertionSort (25 Marks)
Initially, use the **unique alcohol content values** of each wine from the combined red wine and white wine .csv files.

a) Implement a proper **Insertion Sort algorithm** so that all alcohol content values are sorted in ascending order.
b) Calculate the time complexity for sorting the dataset. Does it change if you randomly shuffle the list before sorting? Why or why not?

Problem 3. MergeSort (25 Marks)
Initially, use the **unique alcohol content values** of each wine from the combined red wine and white wine .csv files.

a) Implement a proper **Merge Sort algorithm** to order all alcohol content values.
b) Count the number of merge operations needed to sort the dataset. Does this number change if you randomly shuffle the list before sorting? Why or why not?

Problem 4: QuickSort (25 Marks)
Initially, use the **unique alcohol content values** of each wine from the combined red wine and white wine .csv files.

a) Implement a proper **Quick Sort algorithm** with the following pivot selection strategies, ensuring the list is sorted in ascending order:
    a. Choose the first element as the pivot
    b. Choose the last element as the pivot
    c. Choose a random element as the pivot
    d. Choose the median of the three as the pivot
b) Count the number of comparisons for each pivot strategy required to sort the dataset.
    a. Does the number of comparisons change depending on the pivot strategy?
    b. Which pivot selection strategy works best for this dataset, and why?